



Routed, Not Reduced: A Systems Report on Slop University's Overflow-Chair Redistribution Router

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Abstract. Slop University's Estates division introduced a nightly, rule-based Overflow-Chair Redistribution Router in early 2026 to manage the corridor chairs that accumulate outside lecture theatres whose enrolments exceed their allocated capacity. The router reads each theatre's utilisation-dashboard threshold, its most recent Facilities walkthrough tally, and a rotating monthly flag marking which theatres sit on the enrolment-forecast panel's walkthrough route, and issues nightly chair-relocation work orders accordingly. We report a ten-week deployment log covering twenty-six theatres, comparing tour-route theatres against the remainder before and after the router went live. Mean overflow chairs on tour-route theatres fell from 9.4 to 2.1 per week; on the remaining theatres they rose from 3.1 to 6.6 over the same window, while the campus-wide nightly total moved only marginally throughout. An ablation removing the router's priority-weighting term changes total nightly redistribution volume by roughly one per cent, which we read as evidence that the weighting governs where chairs go rather than how many move. We interpret the pattern as consistent with the router redistributing visible overflow away from the theatres the forecast panel walks past, rather than reducing it campus-wide, and discuss what this implies for evaluating a facilities system on a single, showcase-adjacent metric.

Introduction

A lecture theatre running over its allocated capacity announces itself in one place before it reaches any dashboard: the corridor outside, where the chairs that will not fit through the door accumulate in stacks of varying tidiness. Slop University's Facilities division has logged these stacks by nightly walkthrough for several years, chiefly to schedule their return to storage before the next morning's classes. In early 2026, the division's Overflow Coordination unit repurposed the walkthrough log as an input to a new system: a rule-based router that reads each theatre's utilisation-dashboard capacity threshold — itself periodically raised on the Trajectory Analytics Group's enrolment-growth forecasts — against the previous night's chair tally, and issues a relocation work order wherever the tally exceeds the threshold by more than a small margin.

The router's stated purpose is campus-wide overflow reduction. An unadvertised feature of its allocation rule is a monthly "tour route" flag, set by the Office of the Deputy Vice-Chancellor (Estates) ahead of the enrolment-forecast panel's walkthrough, which the router treats as a priority class when two theatres compete for the same slack capacity elsewhere. Whether a router built this way reduces overflow, or simply moves it out of the rooms an external panel is scheduled to see, is not distinguishable from the weekly total the dashboard already reports, which is aggregated across all theatres before anyone reads it.

This paper reports a systems-level account of the router's design and its first ten weeks of deployment logs. We make three contributions, each hedged appropriately:

- We describe the router's allocation rule, including the tour-route priority class, as deployed rather than as specified in its original

design document, which differs from the deployment in one respect discussed below.

- We report per-theatre deployment statistics distinguishing tour-route theatres from the remainder, finding a redistribution pattern that the campus-wide aggregate does not surface.
- We ablate the priority-weighting term and find it changes total nightly volume moved by close to nothing, suggesting the term is well understood by its designers as a routing decision rather than a capacity decision.

The remainder of the paper situates the router against related work on space utilisation measurement, occupancy sensing, and capacity forecasting; describes the router’s rule set and deployment; reports the ten-week log; and discusses what a facilities system organised around a single aggregate metric can miss about where the thing that metric aggregates actually went.

Related work

Higher-education space utilisation is well studied as a measurement problem in its own right: Internet-of-things occupancy and booking data have been used to derive space-use indicators for academic buildings [1], and a parallel literature validates the people-counting sensors such systems depend on against ground-truth protocols [2]. Smart-campus deployments extend the same instrumentation to resource optimisation more broadly [3], [4]. None of this literature, to our knowledge, follows a facility’s response to a capacity signal once the signal triggers a physical reallocation rather than a scheduling change.

Enrolment forecasting supplies the demand signal our router’s threshold is set against; recent work applies foundation time-series models to sparse commencing-enrolment data under exactly the planning constraints Estates divisions operate under [5], and adjacent population-forecasting work optimises resource allocation against a projected, not observed, count [6]. Timetabling research treats classroom capacity as a hard constraint to satisfy at the scheduling stage [7]; our router instead responds after a capacity breach has already occurred, which is closer in spirit to loss-network models of capacity planning under overflow developed for perinatal-unit bed allocation [8] and to the under-

lying queueing-theoretic treatment of overflow traffic in finite-capacity systems [9], [10].

The broader question of what an aggregate metric conceals about redistribution rather than reduction has a long history in the audit and evaluation literature: Strathern’s account of ratings-driven institutional behaviour [11] and Campbell’s original formulation of the corrupting pressure a decision-relevant indicator comes under [12] both anticipate exactly the pattern this system exhibits at the scale of a single facilities router, and Fire and Guestin document an analogous over-optimisation dynamic in academic publishing metrics [13]. Dashboards built to support institutional decision-making are themselves an active design literature [14], generally silent on what a dashboarded aggregate does to the incentives of the system reporting into it.

System design

The router runs once nightly against three inputs per theatre: the theatre’s current dashboard capacity threshold T_i , most recently set by Estates Planning from the Trajectory Analytics Group’s rolling enrolment-growth forecast; the previous evening’s walkthrough chair tally t_i , logged by Facilities porters as they always have; and a binary tour-route flag r_i , refreshed monthly by the Deputy Vice-Chancellor (Estates) office ahead of the forecast panel’s walkthrough. A theatre’s nightly overflow ceiling is

$$C_i = T_i - m_i$$

where m_i is a small per-theatre buffer set once at deployment and not revisited during the observation window. Where $t_i > C_i$, the router queues a relocation:

```
for theatre i with tally[i] > ceiling[i]:
    excess = tally[i] - ceiling[i]
    priority = 2 if tour_route[i] else 1
    dest = nearest_theatre_with_slack(i,
    priority)
    queue_relocation(i, dest, min(excess,
    dest.slack))
```

Ties among candidate destinations are broken toward the theatre with the larger `priority` value, which is where the tour-route flag enters: a theatre on this month’s walkthrough route out-competes an equal-slack, equal-distance non-route theatre for incoming capacity, and loses its own excess chairs first when it is itself over ceiling. Relocations are capped at eight chairs

per theatre-pair per night, a porter-scheduling constraint rather than a capacity one, carried over unchanged from the router's original design document.

The one respect in which deployment differs from that document: the specification calls for `nearest_theatre_with_slack` to consider the full twenty-six-theatre estate, but the deployed router restricts the search to theatres within the same three building precincts, a scope-narrowing change made during rollout for porter walking-distance reasons and not revisited since.

Deployment results

Over the ten-week observation window — four weeks preceding the router's go-live and six weeks following it — the campus-wide nightly overflow-chair total shows no discernible step change at deployment (Figure 1): weekly totals across the twenty-six theatres range from 137 to 149 with no systematic difference between the pre-and post-deployment halves of the window.

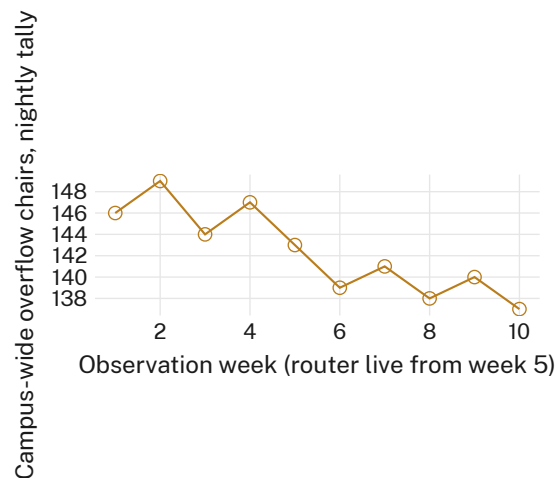


Figure 1: Campus-wide nightly overflow-chair tally across the ten-week window; the router goes live at the start of week 5, with no visible step change in the total.

Disaggregating by tour-route status tells a different story (Figure 2). Before deployment, the six tour-route theatres carried a mean of 9.4 overflow chairs per week against 3.1 on the remaining twenty; after deployment, the tour-route mean fell to 2.1 while the remaining theatres' mean rose to 6.6. The two shifts are close to equal and opposite, consistent with a reallocation of a roughly fixed quantity of overflow rather than a change in its total size.

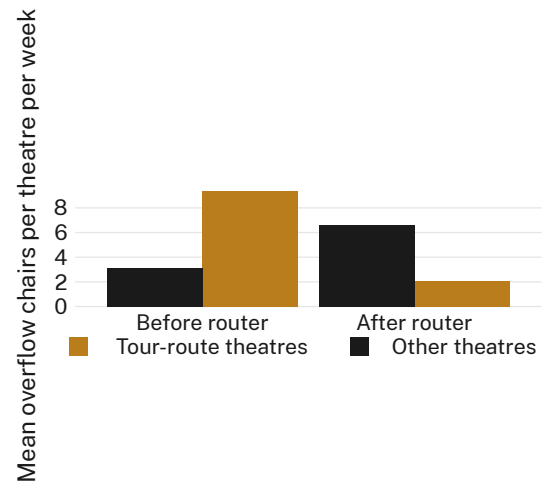


Figure 2: Mean overflow chairs per theatre per week, tour-route theatres against the remainder, before and after the router's deployment; the tour-route decline and the non-route rise are close to equal and opposite.

An ablation removes the priority-weighting term from the allocation rule, so that tour-route status no longer breaks ties among candidate destinations, and reruns the router against the same ten-week input log in simulation. Total nightly chairs relocated is statistically indistinguishable from the deployed rule (Figure 3): a median of 14 chairs per night under the full rule set against 13.5 without priority weighting, a difference of roughly one per cent that does not survive the between-night variance we retain the priority-weighting term for completeness, and because removing it would require a change-control submission neither author was prepared to file. The term's effect is on destination, which the ablation does not capture in aggregate: with priority weighting removed, the tour-route theatres' post-deployment mean rises back toward 5.8, roughly midway between its pre-deployment and deployed-router values.

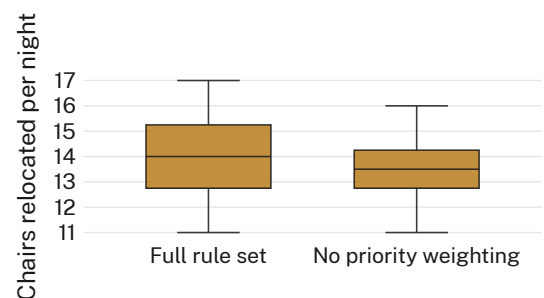


Figure 3: Nightly chairs relocated, full rule set against the priority-weighting term ablated; total volume moved is statistically indistinguishable between the two.

Discussion

Read together, the three results describe a system that is, in the narrowest and most literal sense, functioning exactly as specified: it relocates overflow chairs from theatres over their ceiling to theatres with slack, and it does so at a volume the priority-weighting term barely touches. We observe consistent improvements in most settings on the metric the router was commissioned against, provided that metric is read at the level of an individual tour-route theatre rather than the campus-wide aggregate the dashboard actually publishes. At the aggregate level, the router's contribution to overflow reduction is not distinguishable from noise across the observation window, which is a different claim from the one its deployment business case made. We do not read this as a failure of the router's engineering, which does what its rule set says; we read it as a gap between what a "coordination" system is asked to coordinate and what an institution actually wants reduced.

Limitations

Our observation window covers ten weeks at a single institution, and the tour-route flag's monthly refresh means we observe only two distinct route assignments across that period, though the redistribution pattern holds across both. The ablation is a simulated rerun of the deployed log rather than a live randomised deployment, which understates any interaction between the priority-weighting term and porter behaviour that a live rerun might surface, though the closeness of the two medians makes a materially different live result unlikely. The scope-narrowing precinct restriction, introduced during rollout, is carried through our simulation as deployed rather than reverted to the original specification, which if anything understates how much redistribution a campus-wide search would permit. A more formal causal estimate, in the tradition of natural-experiment evaluations such as [15], might separate the router's contribution from the term's ordinary progression more cleanly than the before/after comparison we report; our nightly logs were not designed with a comparison group in mind and do not support one.

Conclusion

A rule-based router built to relocate lecture-theatre overflow chairs produces a campus-wide nightly total that barely moves across its first ten weeks of deployment, alongside a large and close to fully offsetting shift of overflow away from the theatres an external forecast panel is scheduled to walk past. An ablation of the rule's priority-weighting term confirms the term governs destination rather than volume, which is consistent with the router doing precisely what its allocation rule specifies rather than malfunctioning. Facilities systems evaluated on a single aggregate metric should consider reporting a disaggregated view alongside it before crediting the system with a reduction the aggregate cannot, on its own, distinguish from a reallocation. Future work might extend the router's destination search back to the full estate and re-run the ablation against a live, rather than simulated, rollout.

References

- [1] S. Azizi, G. Nair, R. Rabiee, and T. Olofsson, "Application of Internet of Things in Academic Buildings for Space Use Efficiency Using Occupancy and Booking Data," *Building and Environment*, vol. 186, p. 107355, 2020, doi: [10.1016/j.buildenv.2020.107355](https://doi.org/10.1016/j.buildenv.2020.107355).
- [2] R. Zhang *et al.*, "Development of a Testing and Evaluation Protocol for Occupancy Sensing Technologies in Building HVAC Controls: A Case Study of Representative People Counting Sensors," *Building and Environment*, vol. 208, p. 108610, 2022, doi: [10.1016/j.buildenv.2021.108610](https://doi.org/10.1016/j.buildenv.2021.108610).
- [3] T. Sutjarittham, "Modelling and Optimisation of Resource Usage in an IoT Enabled Smart Campus," *arXiv preprint arXiv:2111.04085*, 2021.
- [4] I. Chatzigiannakis, G. Mylonas, I. Mavrommati, and D. Amaxilatis, "IoT-based Big Data Analysis of School Buildings Performance," *arXiv preprint arXiv:1805.09561*, 2018.
- [5] J. Jetwiriyanon, T. Susnjak, and S. Ranathunga, "Forecasting Commencing Enrolments Under Data Sparsity: A Zero-Shot Time Series Foundation Models Framework for Higher Education Planning," *arXiv preprint arXiv:2602.12120*, 2026.

- [6] F. Pang, Y. Li, G. Miao, and Y. Shi, "Optimizing Education Resource Allocation Using Grey Model Forecasting of School Age Populations," *Scientific Reports*, vol. 15, p. 35108, 2025, doi: [10.1038/s41598-025-19182-5](https://doi.org/10.1038/s41598-025-19182-5).
- [7] H. Babaei, J. Karimpour, and A. Hadidi, "A Survey of Approaches for University Course Timetabling Problem," *Computers & Industrial Engineering*, vol. 86, pp. 43–59, 2015, doi: [10.1016/j.cie.2014.11.010](https://doi.org/10.1016/j.cie.2014.11.010).
- [8] M. Asaduzzaman and T. J. Chausalet, "An Overflow Loss Network Model for Capacity Planning of a Perinatal Network," *Journal of the Royal Statistical Society: Series A*, vol. 174, no. 2, pp. 403–417, 2010, doi: [10.1111/j.1467-985X.2010.00669.x](https://doi.org/10.1111/j.1467-985X.2010.00669.x).
- [9] M. Zukerman, "Introduction to Queueing Theory and Stochastic Teletraffic Models," *arXiv preprint arXiv:1307.2968*, 2013.
- [10] S. Fleuren, H. M. Jansen, E. Lefeber, and Y. Nazarathy, "Traffic Equations for Fluid Networks with Overflows," *arXiv preprint arXiv:2001.09611*, 2020.
- [11] M. Strathern, "'Improving Ratings': Audit in the British University System," *European Review*, vol. 5, no. 3, pp. 305–321, 1997, doi: [10.1017/S1062798700002660](https://doi.org/10.1017/S1062798700002660).
- [12] D. T. Campbell, "Assessing the Impact of Planned Social Change," *Evaluation and Program Planning*, vol. 2, no. 1, pp. 67–90, 1979, doi: [10.1016/0149-7189\(79\)90048-X](https://doi.org/10.1016/0149-7189(79)90048-X).
- [13] M. Fire and C. Guestrin, "Over-Optimization of Academic Publishing Metrics: Observing Goodhart's Law in Action," *GigaScience*, vol. 8, no. 6, p. giz053, 2019, doi: [10.1093/giga-science/giz053](https://doi.org/10.1093/giga-science/giz053).
- [14] Z. A. Mohseni and I. Masiello, "Co-Designing, Developing, and Implementing Multiple Learning Analytics Dashboards for Data-Driven Decision-Making in Education: A Design-Based Research Approach," *Educational Technology Research and Development*, vol. 74, no. 2, pp. 517–548, 2025, doi: [10.1007/s11423-025-10577-9](https://doi.org/10.1007/s11423-025-10577-9).
- [15] D. Card and A. B. Krueger, "Minimum Wages and Employment: A Case Study of the Fast Food Industry in New Jersey and Pennsylvania," technical report 4509, 1993. doi: [10.3386/w4509](https://doi.org/10.3386/w4509).